

Show understanding of the features provided by a Database Management System (DBMS) that address the issues of a file based approach

9618_w25_qp_13_sc_5.d

QUESTION

(d) A Database Management System (DBMS) is used to maintain and manage the database.

Describe **two** ways in which the DBMS can be used to ensure the security of the customer data.

1

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2

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.....

[4]

MARK SCHEME

5(d)	One mark for identification of method one mark for corresponding description max 4 marks • Authentication methods / passwords / biometrics / 2-factor authentication can be implemented • ... which prevents unauthorised access to the customer's data • Access rights / privileges can be set • ... so that only those with correct permissions can read / edit customer's data • Regular backups can be scheduled • ... so that a second copy of the customer's data is available in case of loss/damage • The data can be encrypted • ... so that the customer's data cannot be understood by anyone who gains unauthorized access • Different views can be created • ... so that not everyone can see the customer's data	4
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9618_s25_qp_13_sc_6.d.i

QUESTION

(d) The shop uses a Database Management System (DBMS).

(i) One feature provided by a DBMS is a data dictionary.

The data dictionary stores the attribute names, table names, foreign keys and primary keys in the database.

Identify **three** other items stored in a data dictionary.

1

2

3 [3]

MARK SCHEME

6(d)(i)	1 mark each to max 3 e.g. • Relationships • Views • Data types • Validation rules	3
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QUESTION

(c) A Database Management System (DBMS) provides data security.

- (i) Identify **two** methods the DBMS can use to protect the data in the table `USER` from unauthorised access.

Explain how each method protects the data.

Method 1

Explanation

.....

.....

Method 2

Explanation

.....

.....

[4]

MARK SCHEME

5(c)(i)	1 mark for method 1 mark for matching explanation e.g. • Access rights • appropriate permissions for the table <code>USER</code> are needed to read or edit the data • A password for the database or the <code>USER</code> table • prevents users without the password from accessing the data • Encrypting the database • stops users without the decryption key from decoding / understanding the data • Views • users can be given a view to the database that does not include the data in the table <code>USER</code>	4
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QUESTION

(ii) The DBMS also supports data integrity.

Give **two** ways that a DBMS can support data integrity.

- 1
-
- 2
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[2]

MARK SCHEME

5(c)(ii)	1 mark each to max 2 e.g. • Validation • Enforce referential integrity • Cascade update / delete • Ensuring the database is normalised	2
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QUESTION

(e) A Database Management System (DBMS) supports data integrity.

Explain how a DBMS supports data integrity.

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-

[3]

MARK SCHEME

4(e)	1 mark for each bullet point each (max 3) • Referential integrity is enforced • ... such as cascade update/delete // if the data is changed in one place it is updated in every other place • ... and ensures each foreign key has a corresponding primary key	3
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9618_s24_qp_11_sc_6.b

QUESTION

(b) The company is using a Database Management System (DBMS) to set up the database.

Describe what is meant by the following DBMS features:

Data dictionary

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.....

.....

Logical schema

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.....

[4]

MARK SCHEME

6(b)	1 mark each to max 2 for data dictionary and max 2 for logical schema: Data dictionary: • Data about the data in the database // metadata • Identifies the characteristics of the data that will be stored • Appropriate example e.g. field names, table name, validation rules, data types, primary / foreign keys, relationships etc. Logical schema: • Conceptual design • Platform/database independent overview of the database • Is used to design the physical structure • Appropriate example e.g. Design of entities / E-R diagram / views	4
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QUESTION

- (b) Authentication is one method a Database Management System (DBMS) can use to improve the security of a database.

Describe **other** methods that a DBMS can use to improve the security of a database.

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..... [4]

MARK SCHEME

2(b)	1 mark for each bullet point (max 4) Max 2 if no descriptions • Backup / recovery procedures • ... automatically takes copies of the database and store off site on a regular basis / weekly, etc. • ... so that the data can be recovered if lost • Use of access rights • ... some users are given different access permissions to different tables • ... read/write, read only, full access, etc. • Views • ... different users are able to see different parts of the database • ... only see what users need to see // by example • Record and table locking • ... prevents simultaneous access to data • ... so updates are not lost // data is not overwritten • Encryption • ... the data is turned into ciphertext • ... so it cannot be understood without a decryption key	4
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QUESTION

- 2 A horse riding school uses a database, `Lessons`, to store data about lesson bookings.

This database is created and managed using a Database Management System (DBMS).

- (a) The table contains names and descriptions of DBMS features and tools.

Complete the table by writing down the missing names and descriptions.

Name	Description
Data dictionary	
Query processor	
.....	A model of a database that is not specific to one DBMS.
.....	A software tool that allows the user to create items such as tables, forms and reports.

[4]

MARK SCHEME

2(a)	1 mark for each correct feature or description	4										
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9618_s23_qp_11_sc_2.a.i

QUESTION

2 An organisation uses a database to store data about the types of bird that people have seen.

(a) The database is managed using a Database Management System (DBMS).

(i) State what is meant by a data dictionary **and** give **one** example of an item typically found in a data dictionary.

Definition

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Example

.....

[2]

MARK SCHEME

2(a)(i)	1 mark for definition: - Data about the data in the database // data about the structure of the database // metadata for a database 1 mark for a suitable example Examples: - table names - data types - field names	2
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9618_s23_qp_11_sc_2.a.ii

QUESTION

- (ii) State what is meant by data integrity **and** give **one** example of how this is implemented in a database.

Definition

.....

Example

.....

[2]

MARK SCHEME

2(a)(ii)	1 mark for definition • Methods of making sure the data is consistent 1 mark for example Examples: • Enforcing referential integrity • If data in one table is deleted/edited all tables are updated // cascading update/delete • Validation/verification rules	2
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9618_w22_qp_13_sc_2.b

QUESTION

- (b) A Database Management System (DBMS) has several features.

Identify the feature that describes the relationship between data and its structure.

..... [1]

MARK SCHEME

2(b)	Logical schema	1
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9618_w22_qp_12_sc_5.d.i

QUESTION

(d) (i) Describe, using an example, what is meant by a **data dictionary**.

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..... [2]

MARK SCHEME

5(d)(i)	1 mark for description • stores metadata about the database 1 mark for a correct example For example: • field / attribute names • table name • validation rules • data types • primary keys // foreign keys • relationships	2
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9618_w22_qp_12_sc_5.d.ii

QUESTION

(ii) Describe what is meant by a **logical schema**.

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..... [2]

MARK SCHEME

5(d)(ii)	1 mark for each bullet point (max 2): • the overview of a database structure • models the problem / situation • ... by using methods such as an ER diagram • independent of any particular DBMS	2
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QUESTION

(b) The shop manager uses a Database Management System (DBMS).

Describe the purpose **and** contents of the data dictionary in the DBMS.

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..... [3]

MARK SCHEME

6(b)	1 mark for description of purpose • Stores metadata about the database 1 mark for each example of contents to max 2 e.g. • field / attribute names • table name • validation rules • data types • primary keys // foreign keys • relationships	3
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QUESTION

(b) Describe the ways in which access rights can be used to protect the data in Javier's database from unauthorised access.

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..... [3]

MARK SCHEME

5(b)	1 mark per bullet point • Access rights give managers / himself access to different elements • ... by having different accounts / logins • ... which have different access rights e.g. read only // no access / read / write • Specific <u>views</u> can be assigned to himself and to the managers • ... e.g. managers can only see the data for their own shop(s)	3
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9618_s21_qp_11_sc_7.c

QUESTION

(c) A database has a data dictionary.

Give **three** items that are stored in a data dictionary.

1

2

3 [3]

MARK SCHEME

7(c)	1 mark per item to max 3 • table name • field name // attribute • data type • type of validation • Primary Key • Foreign Key • relationships	3
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Show understanding of how software tools found within a DBMS are used in practice

9618_w25_qp_12_sc_4.d

QUESTION

(d) Describe the purpose of a developer interface in a Database Management System (DBMS).

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.....

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..... [2]

MARK SCHEME

4(d)	1 mark per bullet point, max 2 marks	2
	• To allow the user to create / modify / delete tables // maintain the database • To allow the user to set up / modify relationships • To allow the user to create a form for data input • To allow a user to add tools to a form // for example, drop-down boxes / buttons etc. • To allow a user to design a report (to show the output in an organised manner) • To allow a user to add a menu to enable a choice of different actions • To allow a user to inspect the database contents and metadata • ... by creating and/or running different SQL queries	

QUESTION

- (ii) The DBMS provides a developer interface.

Explain how a database designer can make use of the developer interface.

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..... [3]

MARK SCHEME

6(d)(ii)	1 mark each to max 3 e.g. • To create / modify / delete database objects • To create a form for data input • To add tools to a form • for example, drop-down boxes / buttons etc. • To design a report to show the output in an organised manner • To add a menu to enable users to choose different actions / run different queries	3
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9618_w22_qp_13_sc_2.f

QUESTION

(f) Describe the purpose of a query processor in a DBMS.

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..... [2]

MARK SCHEME

2(f)	1 mark for each bullet point (max 2): - allows the user to enter criteria - searches for the data that meets the entered criteria - organises the results to be displayed to the user	2
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